Lecture 7: Laplace Transforms and Properties - Part 2

CBE 162, Fall 2026

1 Partial Fraction Expansions

We can use Laplace transforms to readily solve linear ODEs as long as the pair $y(t) \Leftrightarrow Y(s)$ appears in the table. What should we do if the expression we derive for $Y(s)$ is not in the table? This is actually a very common occurrence since it is impractical to include every possible case in the limited table. Instead, we need a way to breakdown our expression for $Y(s)$ into the constituent/elementary building blocks that have been included in the table. We provide an overview of a general procedure for doing this in this section, which is referred to as *partial fraction expansion (PFE)*.

1.1 Introductory Example

Let us start with the following motivating example

$$Y(s) = \frac{s+5}{s^2+5s+4} = \frac{s+5}{(s+1)(s+4)}. \tag{1}$$

Notice that this expression *does not* appear in the table due to the $s+5$ term in the numerator. It turns out that we can expand this expression into the sum of two *partial fraction* factors

$$Y(s) = \frac{s+5}{(s+1)(s+4)} = \underbrace{\frac{\alpha_1}{s+1} + \frac{\alpha_2}{s+4}}_{\text{partial fractions}}, \tag{2}$$

where α_1 and α_2 are unknown coefficients that must satisfy the equation in (2). There are many available methods for calculating these coefficients, which we will discuss in more detail after describing the general PFE method. If we use the "equate like coefficients" method, we simply multiply both sides of (2) by the denominator to arrive at

$$s+5 = \alpha_1(s+4) + \alpha_2(s+1), \tag{3a}$$
$$= \alpha_1 s + 4\alpha_1 + \alpha_2 s + \alpha_2, \tag{3b}$$
$$= (\alpha_1+\alpha_2)s + (4\alpha_1+\alpha_2). \tag{3c}$$

Since both sides must be equal, we see that $1 = \alpha_1 + \alpha_2$ and $5 = 4\alpha_1 + \alpha_2$ since the coefficients on the left-hand side $s+5 = (1)s^1 + (5)s^0$ must equal those on the right-hand side. We can solve these two equations for two unknowns by substituting $\alpha_1 = 1 - \alpha_2$ into the second equation

$$5 = 4(1-\alpha_2) + \alpha_2 = 4 - 4\alpha_2 + \alpha_2 = 4 - 3\alpha_2 \quad \rightarrow \quad \alpha_2 = -1/3. \tag{4}$$

Substituting this into the first equation, we get $\alpha_1 = 1 - \alpha_2 = 1 + 1/3 = 4/3$. Thus, the original expression (1) simplifies to

$$Y(s) = \frac{\frac{4}{3}}{s+1} + \frac{-\frac{1}{3}}{s+4}. \tag{5}$$

Since these expressions appear in the provided table, we can directly invert $Y(s)$ to derive $y(t)$:

Start with the definition and the reduced expression

$$y(t) = \mathscr{L}^{-1}[Y(s)] = \mathscr{L}^{-1}\left[\frac{\frac{4}{3}}{s+1} + \frac{-\frac{1}{3}}{s+4}\right] \tag{6a}$$

Take advantage of linearity of the inverse Laplace transform

$$= \frac{4}{3}\mathscr{L}^{-1}\left[\frac{1}{s+1}\right] - \frac{1}{3}\mathscr{L}^{-1}\left[\frac{1}{s+4}\right] \tag{6b}$$

Identify the exponential function as being the correct pair from the table

$$= \frac{4}{3}e^{-t}\gamma(t) - \frac{1}{3}e^{-4t}\gamma(t). \tag{6c}$$

Note: We will often drop the step function $\gamma(t)$ that technically multiplies every time domain function derived from an inverse Laplace transform, as the function must be zero before $t = 0$.

1.2 General Procedure

Let us now state the more general procedure that covers all cases of interest in PFE. To do this, we need a general representation of $Y(s)$ that represents the solution to any linear ODE in the s domain. Let us consider the following general representation

$$Y(s) = \frac{N(s)}{D(s)}, \tag{7}$$

where $N(s)$ is the numerator polynomial and $D(s)$ is the denominator polynomial. For the previous example, $N(s) = s + 5$ and $D(s) = s^2 + 5s + 4$.

Question: Why will $Y(s)$ generally be represented as a ratio of polynomials?

Answer: As shown in the Laplace transform property table, derivatives with respect to the output $y(t)$ or the input $u(t)$ lead to powers of s once $\mathscr{L}[\cdot]$ is applied. Thus, we generally can write out $D(s) = a_n s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \cdots + a_0$ and $N(s) = b_m s^m + b_{m-1}s^{m-1} + b_{m-2}s^{m-2} + \cdots + b_0$.

Question: For $D(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_0$, what is the order of the corresponding ODE model?

Answer: The ODE will be n-th order since there must be derivatives up until $\frac{d^n y}{dt^n}$ appearing in the model due to the term $D(s)Y(s) = a_n s^n Y(s) + \cdots + a_0 Y(s) = N(s)$.

The following is an important fact about the relationship between the numerator $N(s)$ and denominator $D(s)$ polynomials for physically realizable systems. We will prove this result later in the course, but for now just take it as a check that a derived model is not incorrect.

For physical systems, the order of $D(s)$ must be greater than the order of $N(s)$, i.e., $n > m$.

Given (7), we can break down all potential PFEs into three cases related to the characteristics of the denominator polynomial $D(s)$:

- $D(s)$ has real and distinct factors;
- $D(s)$ has real and repeated factors;
- $D(s)$ has complex factors.

As long as we know how to deal with each of these specific cases, we can derive a PFE for any given $Y(s)$. We separately discuss these three cases next, mainly though the use of examples; however, the concepts below can be combined together when $D(s)$ has a mixture of real, complex, and repeated roots. Notice the numerator polynomial $N(s)$ does not play a role, though it will affect the determination of the unknown coefficients.

1.2.1 Real and distinct factors

We saw this case in the illustrative example. We can directly break $Y(s)$ up into a sum of partial fraction terms (one for each root in the denominator). Let us take another example:

$$Y(s) = \frac{1}{s(s+1)(s+2)(s+3)} = \frac{\alpha_1}{s} + \frac{\alpha_2}{s+1} + \frac{\alpha_3}{s+2} + \frac{\alpha_4}{s+3}. \tag{8}$$

The inverse transform is then given by

$$\begin{aligned} y(t) = \mathscr{L}^{-1}[Y(s)] &= \alpha_1 \mathscr{L}^{-1}\left[\frac{1}{s}\right] + \alpha_2 \mathscr{L}^{-1}\left[\frac{1}{s+1}\right] + \alpha_3 \mathscr{L}^{-1}\left[\frac{1}{s+2}\right] + \alpha_4 \mathscr{L}^{-1}\left[\frac{1}{s+3}\right], \\ &= \alpha_1 \underbrace{\gamma(t)}_{\text{can think of as } e^{-0t}} + \alpha_2 e^{-t} + \alpha_3 e^{-2t} + \alpha_4 e^{-3t}. \end{aligned} \tag{9}$$

Key takeaways:

- The solution in the time domain $y(t)$ will always be the sum of exponential terms, one for each distinct factor in $D(s)$.
- The location of the root $s - a$ tells us what behavior will happen. $y(t)$ will decay when $a < 0$ and blow up when $a > 0$.
- It only takes *one positive root* (leading to e^{at}) to result in a blow up. All roots must be negative to have a decaying/stable response.

1.2.2 Real and repeated factors

In this case, the PFE procedure is similar to before, except now we need to include powers up until the order of the repeated root. Suppose that $Y(s)$ has the form

$$Y(s) = \frac{1}{(s+b)^n}, \tag{10}$$

then the appropriate form for the PFE is

$$Y(s) = \frac{1}{(s+b)^n} = \frac{\alpha_1}{s+b} + \frac{\alpha_2}{(s+b)^2} + \cdots + \frac{\alpha_n}{(s+b)^n}. \tag{11}$$

To see how this works when combined with real, distinct roots, let us take another specific example

$$Y(s) = \frac{s+1}{s(s+2)^2} = \frac{\alpha_1}{s+2} + \frac{\alpha_2}{(s+2)^2} + \frac{\alpha_3}{s}, \tag{12}$$

where you see we have combined the terms from the real, repeated root with those of the real, distinct root. It turns out we can also find the inverse transform using the table

$$\begin{aligned} y(t) = \mathscr{L}^{-1}[Y(s)] &= \alpha_1 \mathscr{L}^{-1}\left[\frac{1}{s+2}\right] + \alpha_2 \mathscr{L}^{-1}\left[\frac{1}{(s+2)^2}\right] + \alpha_3 \mathscr{L}^{-1}\left[\frac{1}{s}\right], \\ &= \alpha_1 e^{-2t} + \alpha_2 t e^{-2t} + \alpha_3 \gamma(t). \end{aligned} \tag{13}$$

Question: Does the second term decay as $t \to \infty$?

Answer: Yes, since the exponential decays faster than linear term. You can verify this using L'Hôpital's rule.

Key takeaways:

- When real, repeated roots are present in $D(s)$, include additional terms in PFE with powers up until order of repeated root as shown in (11).
- Combine with other partial fraction factors if real, distinct roots are present.

1.2.3 Complex factors

In this case, the PFE procedure can be performed in exactly the same way as real, distinct roots, we just need to keep track of imaginary parts. However, this process can become quite cumbersome, as illustrated in Figure 2. In Figure 2, we show why these complex roots lead to oscillations, which simply mean $\sin(\omega_0 t)$ and $\cos(\omega_0 t)$ terms that either appear by themselves or multiplied by decaying exponential terms. Looking at the Laplace transform pairs table, we see several terms with quadratic denominators in s that are explicitly included. Therefore, it is often preferred to convert complex roots to this type of *standard form* using a procedure known as *completing the square*. Let us see how this procedure works. Suppose that $Y(s)$ has the form

$$Y(s) = \frac{1}{as^2 + bs + c} = \frac{1}{(s - r_1)(s - r_2)}, \tag{14}$$

for constants a, b, c. Recall that the roots of the denominator polynomial can be found with the quadratic formula in this case: $r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. These roots will be complex whenever $b^2 < 4ac$, such that completing the square gives us

$$Y(s) = \frac{1}{as^2 + bs + c} = \frac{\frac{1}{a}}{\left(s + \frac{b}{2a}\right)^2 + \left[\frac{c}{a} - \left(\frac{b}{2a}\right)^2\right]}. \tag{15}$$

where the term $\frac{c}{a} - \left(\frac{b}{2a}\right)^2 > 0$ whenever we have complex roots that lead to oscillations. Therefore, whenever we have complex roots that are not directly in the form of one of the entries in the table, we can apply (15) to put the partial fraction in the desired form.

To see how this works when combined with the previously considered cases, let us take a look at another specific example

$$Y(s) = \frac{1}{(s^2 + 1)(s + 3)^2} = \frac{\alpha_1 s + \alpha_2}{s^2 + 1} + \frac{\alpha_3}{s + 3} + \frac{\alpha_4}{(s + 3)^2}, \tag{16}$$

where we have a combination of complex roots and real, repeated roots. Notice how we have modified the numerator in the quadratic term to include two unknown coefficients $\alpha_1 s + \alpha_2$ – you can prove that you need this form by combining the complex partial fractions $\frac{\alpha_1'}{s+j} + \frac{\alpha_2'}{s-j}$ into a single factor. It turns out we can again find the inverse transform using the table

$$\begin{aligned} y(t) = \mathscr{L}^{-1}[Y(s)] &= \alpha_1 \mathscr{L}^{-1}\left[\frac{s}{s^2 + 1}\right] + \alpha_2 \mathscr{L}^{-1}\left[\frac{1}{s^2 + 1}\right] + \alpha_3 \mathscr{L}^{-1}\left[\frac{1}{s + 3}\right] + \alpha_4 \mathscr{L}^{-1}\left[\frac{1}{(s + 3)^2}\right], \\ &= \alpha_1 \cos(t) + \alpha_2 \sin(t) + \alpha_3 e^{-3t} + \alpha_4 t e^{-3t}. \end{aligned} \tag{17}$$

Key takeaways:

- Complex roots in $D(s)$ results in oscillations (sine and/or cosine) in the time domain.
- When complex roots are present in $D(s)$, it is best to put the quadratic equation into standard form by completing the square as shown in (15).
- Combine with other partial fraction factors if real, distinct or real, repeated roots are present.

1.3 How to Find Coefficients?

In the previous section, we saw how to construct the PFE structure for the three different types of roots that may occur in $D(s)$. However, we left these expressions in terms of unknown coefficients $\alpha_1, \alpha_2, \cdots$, which must be determined to derive the exact expression for $y(t)$. There are several ways of going about computing these coefficients. We introduce three potential methods below.

1.3.1 Method 1: Equate like coefficients

This is the most general method, as it works in all cases (for real+distinct, real+repeated, and complex roots). However, it often requires a significant amount of algebra and solving a set of coupled equations for the unknown coefficients.

1.3.2 Method 2: Choose s values and evaluate both sides at this s

Since the derived equation in a PFE, e.g., $\frac{s+5}{(s+1)(s+4)} = \frac{\alpha_1}{s+1} + \frac{\alpha_2}{s+4}$ must hold for all values of s, we can choose to evaluate both sides at specific values of s to construct consistent equations for α_1 and α_2.

Since we need the same number of equations as unknowns, we must evaluate at 2 values in this case. Let us randomly pick $s = 0$ and $s = 1$. For $s = 0$, we get the following equation

$$\frac{5}{4} = \alpha_1 + \frac{1}{4}\alpha_2 \quad \rightarrow \quad 5 = 4\alpha_1 + \alpha_2. \tag{18}$$

For $s = 1$, we get another (different) equation

$$\frac{6}{(2)(5)} = \frac{1}{2}\alpha_1 + \frac{1}{5}\alpha_2 \quad \rightarrow \quad \frac{6}{5} = \alpha_1 + \frac{2}{5}\alpha_2. \tag{19}$$

Now we need to solve these equations. Let us rearrange the second one $\alpha_1 = \frac{6}{5} - \frac{2}{5}\alpha_2$ and substitute into first

$$\begin{aligned} 5 &= 4\left(\frac{6}{5} - \frac{2}{5}\alpha_2\right) + \alpha_2, \\ \frac{25}{5} &= \frac{24}{5} - \frac{8}{5}\alpha_2 + \frac{5}{5}\alpha_2, \\ \frac{1}{5} &= -\frac{3}{5}\alpha_2, \\ -\frac{1}{3} &= \alpha_2 \end{aligned} \tag{20}$$

Substituting this back into the first, we get $\alpha_1 = \frac{6}{5} - \frac{2}{5}\left(-\frac{1}{3}\right) = \frac{18}{15} + \frac{2}{15} = \frac{20}{15} = \frac{4}{3}$. We see that the process yields the same result as the "equate like coefficients" method, which should be the case. However, notice that this method also took quite a bit of algebra. It turns out that the specific choices of s selected matter quite a lot when it comes to the structure of the PFE. This leads us to Method 3, described next.

1.3.3 Method 3: Multiply by $D(s)$ and set s equal to the value so that terms cancel

The fastest approach for computing the coefficients is known as the "Heaviside expansion" method. The procedure works great for real, distinct roots (but must be careful when repeated or complex roots appear). Let us assume the following expansion holds after breaking $Y(s)$ into its partial fraction components

$$Y(s) = \frac{N(s)}{D(s)} = \sum_{i=1}^{n} \frac{\alpha_i}{s + b_i}, \tag{21}$$

where we have a n-th order denominator polynomial (with n real, distinct roots) and n unknown coefficients. The i-th coefficient can be determined by multiplying both sides by $(s + b_i)$ and then evaluating at $s = -b_i$. Notice that this procedure will eliminate all terms except for $\frac{\alpha_i}{s+b_i}(s + b_i) = \alpha_i$ such that

$$\alpha_i = \left[(s + b_i)\frac{N(s)}{D(s)}\right]_{s=-b_i}. \tag{22}$$

To see this method in action, we will also apply it to our illustrative example $\frac{s+5}{(s+1)(s+4)} = \frac{\alpha_1}{s+1} + \frac{\alpha_2}{s+4}$. First, we need to check that we have real, distinct roots – we can see that our roots are at -1 and -4 such that we

should not encounter any issues. Using (22), we can immediately write down separate expressions for our two unknown coefficients

$$\alpha_1 = \left[(s+1)\frac{(s+5)}{(s+1)(s+4)}\right]_{s=-1} = \left[\frac{s+5}{s+4}\right]_{s=-1} = \frac{-1+5}{-1+4} = \frac{4}{3}, \tag{23}$$

$$\alpha_2 = \left[(s+4)\frac{(s+5)}{(s+1)(s+4)}\right]_{s=-4} = \left[\frac{s+5}{s+1}\right]_{s=-4} = \frac{-4+5}{-4+1} = -\frac{1}{3}, \tag{24}$$

We see that this process yields the same result as the previous two methods in significantly less time. This is especially useful for higher-order expressions that have more coefficients.

Question: What happens when we apply the "Heaviside expansion" method for real, repeated roots?

Answer: We will get a divide by 0 error for some terms. Take, for example, $\frac{s+1}{s(s+2)^2} = \frac{\alpha_1}{s} + \frac{\alpha_2}{s+2} + \frac{\alpha_3}{(s+2)^2}$. If we multiply by $s+2$, then the left-hand side will be $\frac{s+1}{s(s+2)}$ so we cannot evaluate at $s=-2$ since this will lead to a divide by zero factor. However, we can still use this method for the other terms. The first term proceeds using (22) exactly, i.e., $\alpha_1 = \left[\frac{s+1}{(s+2)^2}\right]_{s=0} = \frac{1}{4}$. The third term requires us to multiply by $(s+2)^2$, which leads to $\alpha_3 = \left[\frac{s+1}{s}\right]_{s=-2} = \frac{-2+1}{-2} = \frac{1}{2}$. To get the final missing term, we can use Method 2 and simply evaluate at some s value. This procedure simplifies since we already known α_1 and α_3. Let us pick $s=1$ such that we get the following equation

$$\begin{aligned}
\frac{2}{9} &= \frac{\alpha_1}{1} + \frac{\alpha_2}{3} + \frac{\alpha_3}{9}, \\
\frac{2}{9} &= \frac{1}{4} + \frac{\alpha_2}{3} + \frac{1}{18}, \\
\frac{8}{36} &= \frac{9}{36} + \frac{12}{36}\alpha_2 + \frac{2}{36}, \\
-\frac{3}{36} &= \frac{12}{36}\alpha_2, \\
-\frac{1}{4} &= \alpha_2.
\end{aligned} \tag{25}$$

Of course, we could have directly used Method 1 or 2 above on this problem to find $\alpha_1, \alpha_2, \alpha_3$, which should yield the same result with a bit more math.

2 Some Important Laplace Transform Properties

The main value of the Laplace transform is its ability to act on function operators such as the derivative and integral of a function. In this section, we discuss three important properties of Laplace transforms that can be useful in various situations.

2.1 Final Value Theorem

We can use $F(s)$ to look at the asymptotic values of the function $f(t)$ in the time domain. This is referred to as the final value theorem (FVT) and is summarized below.

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0}\left[sF(s)\right]; \quad \text{*if and only if the limit } \lim_{t\to\infty} f(t) \text{ exists} \tag{26}$$

You can prove this result from the derivative property of Laplace transforms

$$\mathscr{L}\left[\frac{df}{dt}\right] = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt = sF(s) - f(0^-). \tag{27}$$

Taking the limit as $s \to 0$ and assuming that $\frac{df}{dt}$ is continuous, we arrive at

$$\int_{0^-}^{\infty} \frac{df}{dt} dt = f(\infty) - f(0^-) = \lim_{s \to 0}[sF(s)] - f(0^-), \tag{28}$$

which directly leads to the FVT. One important restriction is that *the limit* $\lim_{t\to\infty} f(t)$ *must exist and be finite in order to apply FVT*. We need all roots of $D(s)$ in $Y(s) = N(s)/D(s)$ to have non-positive real parts to ensure the limit exists. Note that $F(s)$ can only have up to one 0 root (i.e., one factor of s in $D(s)$) since this will cancel when taking the product $sF(s)$.

FVT Example 1: What is the asymptotic value of $f(t)$ given $F(s) = \frac{1}{s+5}$? We can apply the FVT since the root is -5 (leads to decaying exponential) such that $f(\infty) = \lim_{s\to 0}\left[\frac{s}{s+5}\right] = 0$.

FVT Example 2: What is the asymptotic value of $f(t)$ given $F(s) = \frac{1}{s-5}$? We cannot apply the FVT in this case since the root is 5, which leads to a blowing up exponential. Note the limit $\lim_{s\to 0}[sF(s)]$ still exits in this case, implying we must always check the roots of $D(s)$ before applying the limit in the s domain.

2.2 Initial Value Theorem

Similarly to the FVT, we can also use $F(s)$ to analyze the behavior of $f(t)$ towards $t \to 0^+$ (time approaches zero from the right side). This is referred to as the initial value theorem (IVT) and is summarized below.

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty}[sF(s)] \tag{29}$$

The proof follows the same steps as the FVT, except taking the limit as $s \to \infty$ and recognizing all terms in the integral vanish except for $t = 0$.

IVT Example: What is the value of $f(0^+)$ given $F(s) = \frac{5}{s}$? We have no restrictions on the IVT, so we can directly apply the limit $f(0^+) = \lim_{s\to\infty}\left[s\frac{5}{s}\right] = 5$. This makes sense because $F(s)$ corresponds to a step change of magnitude 5, so the value should be the same for all $t \geq 0^+$.

2.3 Real Translation Theorem

Time delays play an important part in process control, as they often occur when we consider the transport time required for a fluid to flow through a pipe. We illustrate the effect of a time delayed version of a function $f(t)$, denoted by $f_d(t)$, in Figure 1.

From the figure, we can see that $f_d(t) = 0$ before the time of the delay $t < \theta$, which we often denote by θ. For $t \geq \theta$, we must have $f_d(t) = f(t-\theta)$ as the delayed function equals the original function θ time units in the past. We can more succinctly write out this relationship using a step function as follows

$$f_d(t) = f(t-\theta)\gamma(t-\theta). \tag{30}$$

Similarly to how we dealt with the derivative operator in the time domain, we are able to derive an explicit relationship for how delays in the time domain translate to modifications in the s domain. This is referred to as the *Real Translation Theorem* and is summarized as follows

$$\mathscr{L}\left[f(t-\theta)\gamma(t-\theta)\right] = e^{-s\theta}F(s); \quad {}^*F(s) = \mathscr{L}[f(t)] \text{ corresponds to } f(t), \text{ which is not delayed} \tag{31}$$

We see that a delay of θ time units results in a $e^{-s\theta}$ factor in front of $F(s)$. Up until this point, we have only dealt with polynomial functions in s, so you may be asking: How do I deal with $e^{-s\theta}$? Since this only influences the transformation back into the time domain, it actually only adds 1 extra step to the standard PFE procedure discussed in the previous section. Let us explicitly define the delayed Laplace transform

$$F_d(s) = e^{-s\theta}F(s). \tag{32}$$

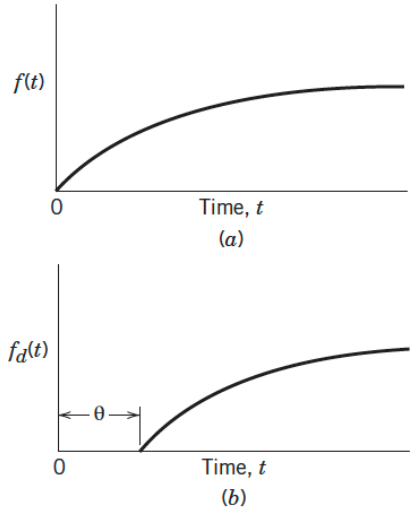


Figure 1: A time domain function with and without time delay. (a) Shows the original function $f(t)$ without any delay; (b) Shows the function delayed by θ time units $f_d(t) = f(t-\theta)\gamma(t-\theta))$.

To derive the time domain expression $f_d(t)$, you follow these two steps:

1. Start by inverting $F(s)$ using PFE to find $f(t)$.
2. Define $f_d(t) = f(t-\theta)\gamma(t-\theta)$, i.e., replace the argument t, wherever it appears in $f(t)$, by the expression $(t-\theta)$. Remember to multiply the resulting function $f(t-\theta)$ by the step function $\gamma(t-\theta)$ to zero out the behavior before the delay.

Real Translation Theorem Example: What is the time domain function $y(t)$ that corresponds to the Laplace transform $Y(s) = \frac{e^{-8s}}{s+2}$? We start by isolating the polynomial part by defining $Y(s) = e^{-8s}F(s)$ where $F(s) = \frac{1}{s+2}$. From the Laplace transform pair table, we see that $f(t) = e^{-2t} \Leftrightarrow F(s) = \frac{1}{s+2}$. Our final step is to apply the Real Translation Theorem with a delay of $\theta = 8$ time units such that $y(t) = f(t-8)\gamma(t-8)$. Substituting in for the correct argument, we arrive at the final answer $y(t) = e^{-2(t-8)}\gamma(t-8)$.

$$Y(s) = \frac{1}{s^2+s+1} = \frac{1}{(s-r_1)(s-r_2)}, \quad r_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{1}{\left(s+\left(\frac{1}{2}-\frac{\sqrt{3}}{2}j\right)\right)\left(s+\left(\frac{1}{2}+\frac{\sqrt{3}}{2}j\right)\right)} \qquad = \frac{-1 \pm \sqrt{1^2-4\cdot 1\cdot 1}}{2\cdot 1} \qquad \sqrt{-1}$$

$$= \frac{-1 \pm \sqrt{-3}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}j$$

Let's look at a simpler example to see the effect of magnary roots $(j=\sqrt{-1})$

$$Y(s) = \frac{1}{s^2+1} = \frac{1}{(s-j)(s+j)} = \frac{\alpha_1}{s-j} + \frac{\alpha_2}{s+j}$$

Solve for α_1, α_2 using PFE $\rightarrow$ $\alpha_1 = -1/2\,j$, $\alpha_2 = 1/2\,j$

$$= \frac{-1/2\,j}{s-j} + \frac{1/2\,j}{s+j}$$

$$y(t) = \mathcal{L}^{-1}[Y(s)] = -1/2\,j\,\mathcal{L}^{-1}\left[\frac{1}{s-j}\right] + 1/2\,j\,\mathcal{L}^{-1}\left[\frac{1}{s+j}\right]$$

from table

$$= -\frac{1}{2}j\,e^{jt} + \frac{1}{2}j\,e^{-jt}$$

Euler's formula $e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$

$$= -\frac{1}{2}j\left[\cos(t) + j\sin(t)\right] + \frac{1}{2}j\left[\cos(-t) + j\sin(-t)\right]$$

Cosine is even $(\cos(x) = \cos(-x))$; Sine is odd $(\sin(x) = -\sin(x))$

$$= -\frac{1}{2}j\left[\cancel{\cos(t)} + j\sin(t)\right] + \frac{1}{2}j\left[\cancel{\cos(t)} - j\sin(t)\right]$$

$$= -\frac{1}{2}j^2\cancel{2}\sin(t) = \boxed{\sin(t)}$$

$j^2 = -1$

* complex roots result in oscillations

Figure 2: Example of partial fraction expansion with complex roots, highlighting why oscillations occur.